

Module IIPARTIAL DIFFERENTIATIONProblems

Q) If $u = x^3 + y^3 - 3axy$ show that $y \frac{\partial^2 u}{\partial x^2} = x \frac{\partial^2 u}{\partial y^2}$

$$u = x^3 + y^3 - 3axy$$

$$\frac{\partial u}{\partial x} = 3x^2 - 3ay$$

$$\frac{\partial u}{\partial y} = 3y^2 - 3ax$$

$$\frac{\partial^2 u}{\partial x^2} = 6x, \quad \frac{\partial^2 u}{\partial y^2} = 6y$$

$$y \frac{\partial^2 u}{\partial x^2} = 6xy, \quad x \frac{\partial^2 u}{\partial y^2} = 6xy$$

$$\therefore y \frac{\partial^2 u}{\partial x^2} = x \frac{\partial^2 u}{\partial y^2}$$

(2) If $u = \log(x^3 + y^3 + z^3 - 3xyz)$

Prove that (i) $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}$

(ii) $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = \frac{-9}{(x+y+z)^2}$

$$u = \log(x^3 + y^3 + z^3 - 3xyz)$$

$$\frac{\partial u}{\partial x} = \frac{1}{x^3 + y^3 + z^3 - 3xyz} (3x^2 - 3yz) = \frac{3(x^2 - yz)}{x^3 + y^3 + z^3 - 3xyz}$$

$$\frac{\partial u}{\partial y} = \frac{3(y^2 - zx)}{x^3 + y^3 + z^3 - 3xyz}, \quad \frac{\partial u}{\partial z} = \frac{3(z^2 - xy)}{x^3 + y^3 + z^3 - 3xyz}$$

$$\therefore \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3(x^2 - yz) + 3(y^2 - zx) + 3(z^2 - xy)}{x^3 + y^3 + z^3 - 3xyz}$$

$$= \frac{3(x^2 + y^2 + z^2 - xy - yz - zx)}{x^3 + y^3 + z^3 - 3xyz}$$

$$= \frac{3(x^2 + y^2 + z^2 - xy - yz - zx)}{(x+y+z)(x^2 + y^2 + z^2 - xy - yz - zx)}$$

$$= \frac{3}{x+y+z}$$

$$\begin{aligned}
 &= \frac{3}{x+y+z} \\
 \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u &= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right) \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right) u \\
 &= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right) \left(\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}\right) \\
 &= \left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right) \left(\frac{3}{x+y+z}\right) \\
 &= \frac{-3}{(x+y+z)^2} + \frac{-3}{(x+y+z)^2} + \frac{-3}{(x+y+z)^2} \\
 &= \frac{-9}{(x+y+z)^2}
 \end{aligned}$$

(3) If $u = x^2 \tan^{-1}\left(\frac{y}{x}\right) - y^2 \tan^{-1}\left(\frac{x}{y}\right)$

Prove that $\frac{\partial^2 u}{\partial x \partial y} = \frac{x^2 - y^2}{x^2 + y^2}$

$$\begin{aligned}
 u &= x^2 \tan^{-1}\left(\frac{y}{x}\right) - y^2 \tan^{-1}\left(\frac{x}{y}\right) \\
 \frac{\partial u}{\partial y} &= x^2 \cdot \frac{1}{\left(1 + \frac{y^2}{x^2}\right)} \left(\frac{1}{x}\right) - \left[y^2 \cdot \frac{1}{\left(1 + \frac{x^2}{y^2}\right)} \left(\frac{-x}{y^2}\right) + 2y \tan^{-1}\left(\frac{x}{y}\right) \right]
 \end{aligned}$$

$$\begin{aligned}
 &= x^2 \cdot \frac{x^2}{x^2 + y^2} \left(\frac{1}{x}\right) + y^2 \left(\frac{y^2}{x^2 + y^2}\right) \left(\frac{-x}{y^2}\right) \\
 &\quad + 2y \tan^{-1}\left(\frac{x}{y}\right)
 \end{aligned}$$

$$= \frac{x^3}{x^2 + y^2} + \frac{-xy^2}{x^2 + y^2} - 2y \tan^{-1}\left(\frac{x}{y}\right)$$

$$= \frac{x(x^2 - y^2)}{x^2 + y^2} - 2y \tan^{-1}\left(\frac{x}{y}\right)$$

$$= x - 2y \tan^{-1}\left(\frac{x}{y}\right)$$

$$\frac{\partial^2 u}{\partial x \partial y} = 1 - 2y \cdot \frac{1}{1 + \frac{x^2}{y^2}} \left(\frac{1}{y}\right) = 1 - \frac{2y(y^2)}{x^2 + y^2} \cdot \frac{1}{y}$$

$$= \frac{1 - 2y^2}{x^2 + y^2} = \frac{x^2 + y^2 - 2y^2}{x^2 + y^2}$$

$$= \frac{x^2 - y^2}{x^2 + y^2}$$

(4) If $u = \log(\tan x + \tan y + \tan z)$

P.T. $\sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} + \sin 2z \frac{\partial u}{\partial z} = 2$

$$u = \log(\tan x + \tan y + \tan z)$$

$$\frac{\partial u}{\partial x} = \frac{1}{\tan x + \tan y + \tan z} \cdot \sec^2 x$$

$$\text{ii) } \frac{\partial u}{\partial y} = \frac{\sec^2 y}{\tan x + \tan y + \tan z}, \quad \frac{\partial u}{\partial z} = \frac{\sec^2 z}{\tan x + \tan y + \tan z}$$

and $\frac{\partial u}{\partial z} = \frac{\sec^2 z}{\tan x + \tan y + \tan z}$

$$\sin 2x \frac{\partial u}{\partial x} = 2 \cdot \sin x \cos x \cdot \frac{1}{\tan x + \tan y + \tan z} \left(\frac{1}{\cos^2 x} \right)$$

$$= \frac{2 \tan x}{\tan x + \tan y + \tan z}$$

$$\sin 2y \frac{\partial u}{\partial y} = \frac{2 \tan y}{\tan x + \tan y + \tan z}$$

$$\sin 2z \frac{\partial u}{\partial z} = \frac{2 \tan z}{\tan x + \tan y + \tan z}$$

$$\sin 2x \frac{\partial u}{\partial x} + \sin 2y \frac{\partial u}{\partial y} + \sin 2z \frac{\partial u}{\partial z}$$

$$= \frac{2(\tan x + \tan y + \tan z)}{\tan x + \tan y + \tan z} = \underline{\underline{2}}$$

(5) If $v = (x^2 + y^2 + z^2)^{-1/2}$ P.T.

$$\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} = 0$$

$$v = (x^2 + y^2 + z^2)^{-1/2} = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$$

put $r^2 = x^2 + y^2 + z^2$

Diff. w. r. to x $2r \frac{\partial r}{\partial x} = 2x \quad \therefore \frac{\partial r}{\partial x} = \frac{x}{r}$

likewise $\frac{\partial r}{\partial y} = \frac{y}{r}, \quad \frac{\partial r}{\partial z} = \frac{z}{r}$

Now $v = \frac{1}{r} \quad \therefore \frac{\partial v}{\partial x} = -\frac{1}{r^2} \frac{\partial r}{\partial x}$

i. $\frac{\partial v}{\partial x} = -\frac{1}{r^2} \cdot \frac{x}{r} = -\frac{x}{r^3}$

$$\begin{aligned} \frac{\partial^2 v}{\partial x^2} &= \frac{\partial}{\partial x} \left(-\frac{x}{r^3} \right) = -\frac{[r^3 \cdot 1 - x \cdot 3r^2 \frac{\partial r}{\partial x}]}{r^6} \\ &= -\frac{[r^3 - 3r^2 \frac{x}{r}]}{r^6} = -\frac{r[r^2 - 3x^2]}{r^6} \end{aligned}$$

$\therefore \frac{\partial^2 v}{\partial x^2} = \frac{r^2 - 3x^2}{r^5}, \quad \text{likewise } \frac{\partial^2 v}{\partial y^2} = \frac{r^2 - 3y^2}{r^5}$

and $\frac{\partial^2 v}{\partial z^2} = \frac{r^2 - 3z^2}{r^5}$

$$\begin{aligned} \text{ii. } \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} &= \frac{3r^2 - 3(x^2 + y^2 + z^2)}{r^5} \\ &= \frac{3r^2 - 3r^2}{r^5} = 0 \end{aligned}$$

Assignment problems.

(1) If $u = \log \left(\frac{x^2 + y^2}{xy} \right)$ P.T. $\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$

(2) If $u = \log r$ where $r^2 = (x-a)^2 + (y-b)^2$

P.T. $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

(3) $u = e^{xyz}$ find $\frac{\partial^3 u}{\partial x \partial y \partial z}$

Total differential Coefft.

If $u = f(x, y)$ where x and y are functions of a third variable t then the total derivative

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt}$$

Note 1 If $u = f(x, y, z)$ where x, y, z are functions of t then the total derivative

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt} + \frac{\partial u}{\partial z} \cdot \frac{dz}{dt}$$

Note 2 If $z = f(x, y)$ where y is a function of x then

$$\frac{dz}{dx} = \frac{\partial z}{\partial x} + \frac{\partial z}{\partial y} \frac{dy}{dx}$$

Derivative of an implicit function

Consider the function $f(x, y) = c$ where c is a constant

Now Let $u = f(x, y)$

by note 2,
$$\frac{du}{dx} = \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \frac{dy}{dx}$$

Since $f(x, y) = c$, $u = c \quad \therefore \frac{du}{dx} = 0$

$$\therefore \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} \frac{dy}{dx} = 0$$

$$\Rightarrow \frac{\partial u}{\partial y} \frac{dy}{dx} = -\frac{\partial u}{\partial x}$$

$$\therefore \frac{dy}{dx} = -\frac{\partial u / \partial x}{\partial u / \partial y}$$

$$\therefore \frac{dy}{dx} = -\frac{u_x}{u_y} \quad \text{where } u_x = \frac{\partial u}{\partial x} \text{ and } u_y = \frac{\partial u}{\partial y}$$

(1) If $ax^2 + 2hxy + by^2 = 0$ find $\frac{dy}{dx}$

Let $u = ax^2 + 2hxy + by^2 = 0$

$$u_x = \frac{\partial u}{\partial x} = 2ax + 2hy$$

$$u_y = \frac{\partial u}{\partial y} = 2hx + 2by$$

$$\begin{aligned} \frac{dy}{dx} &= -\frac{u_x}{u_y} = -\frac{2(ax+hy)}{2(hx+by)} \\ &= -\frac{(ax+hy)}{hx+by} \end{aligned}$$

(2) Find $\frac{dy}{dx}$ if $x^y = y^x$

$$u = x^y - y^x = 0$$

$$\frac{\partial u}{\partial x} = yx^{y-1} - y^x \cdot \log y$$

$$\frac{\partial u}{\partial y} = x^y \cdot \log x - xy^{x-1}$$

$$\frac{dy}{dx} = -\frac{\frac{\partial u}{\partial x}}{\frac{\partial u}{\partial y}}$$

$$= -\frac{(yx^{y-1} - y^x \log y)}{(x^y \log x - xy^{x-1})}$$

(3) If $u = x^5 y^4$ where $x = t^2$, $y = t^3$

Find $\frac{du}{dt}$

$$\frac{du}{dt} = \frac{\partial u}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial u}{\partial y} \cdot \frac{dy}{dt}$$

$$= 5x^4 y^4 \cdot (2t) + 4x^5 y^3 \cdot (3t^2)$$

$$= 10x^4 y^4 t + 12x^5 y^3 t^3$$

$$= 10(t^2)^4 (t^3)^4 t + 12(t^2)^5 (t^3)^3 t^3$$

$$= 10t^{21} + 12t^{21}$$

$$= 22t^{21}$$